Project Deliverable D:

Conceptual Design

Group 16: The Sample Snatchers

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Abstract

This document analyzes the detailed process by which a set of conceptual designs have been developed for the problem statement:

“Operators who service nuclear reactors need a device to collect metal samples from pressure tubes while minimizing exposure to contaminated material. The solution should be compact and completely removable from the pressure tube, and it should also be able to report the stage of operation within a limited timeframe.”

The objective is to create a compact, removable device that minimizes operator exposure to contaminated material while effectively extracting and containing the sample from the pressure tube. Based on previous benchmarking and the list of prioritized design criteria, the team identified four key subsystems – movement, sampling, containment, and failsafe – and generated multiple ideations for each subsystem. These subsystem ideations have been refined and integrated into four fully functional global concepts, which were further conceptualized. A selection matrix evaluates each concept against the design criteria to establish the “best” generated concept.

As a result, the *Pneumatically Deployable Sample Tool* was speculatively the optimal solution, incorporating an inflatable tube for movement, a mechanical arm for sampling, a clamshell containment system, and a self-integrated failsafe for retrieval.

This document details the development of ideations supported by the respective subsystems, which are combined to create various conceptual designs. Justification of the final concept selection process is defined, which will provide a foundation for further development, including prototyping, cost estimation, and validation through client feedback.

1.0 Introduction

1.1 Objective

This document highlights the Ideation process, in which ideas are generated for subsystems with the intent to solve the defined problem statement. The generated ideas are combined into more defined concepts that will be evaluated against the criteria for the design developed in the previous deliverable, Deliverable C.

1.2 Design Criteria

The following design criteria are established to guide the conceptual development of the sample collection device, ensuring functionality, safety, and feasibility. Concepts will combine subsystem ideations to cover the following design criteria:

1. Movement: The device should be able to traverse to and from the sample location consistently, located fifteen feet in the pressure tube.
2. Sample Containment: The sample must be securely enclosed to prevent exposure of contaminated material to the operator. The containment system should minimize the risk of leakage or accidental release.
3. Removability: The device must be fully retrievable from the fifteen-foot pressure. The design should be compact and modular to ensure smooth development and extraction.
4. Sampling Capability: The device must extract a sample size between 30 mg - 80 mg. The sampling mechanism should minimize damage to the pressure tube.
5. Process Tracking: The system should be able to report its operational stage to assist operators in monitoring progress.
6. Modularity and Weight: The device should be lightweight so an operator can move the device to and from the vault.
7. Power source: The device should include all necessary components for the operator. The power source should be implemented.

8. Operational Speed: The device should complete the sampling and retrieval process efficiently to reduce exposure time in radiation-prone environments.
9. Novelty and Innovation: The device should present a unique and innovative approach, incorporating creative engineering solutions.

2.0 Subsystem Concept Generation

2.1 Identification of Subsystems

The team has identified four key subsystems that the final solution must incorporate. These subsystems revolve around the requirements for retrieving and safely containing the sample, building on previous technical and user benchmarking and design criteria derived from user needs.

The subsystems are as follows:

Subsystem 1: Movement - Navigating to and from the sample site inside the fifteen-foot tube.

This subsystem is developed to meet the design criteria of maneuvering through the tube to reach the sample site, located 15 feet inside the pressure tube. The requirements for this subsystem include a method of travelling to and from the sample site at the tube's inlet.

Subsystem 2: Sampling - The method in which sampling is conducted.

This subsystem was developed to meet the design criteria of obtaining the sample from the inner walls of the pressure tube, where the sample size is varied (30 mg—80 mg). It mainly evaluates the methodologies used for sample collection.

Subsystem 3: Containment - Ensuring the sampled material doesn't contact the operator.

This subsystem is developed to meet the design criteria emphasizing safety, including containing the sample while limiting exposure. It explores methods of obtaining the sample without direct contact with the operator.

Subsystem 4: Failsafe - The ability to retrieve the entire device from the tube.

This subsystem is developed to meet the design criteria by incorporating a fail-safe mechanism that ensures the entire device can be removed from the tube. It also considers the integrity of the pressure tube and device.

2.2 Idea Generation for Each Subsystem

Subsystem 1: Movement

The Inflatable Pipe Navigator allows a tool to reach the exact location on the desired sample site in the pressure tube by inflating a cylindrical material to the length of the sample site.

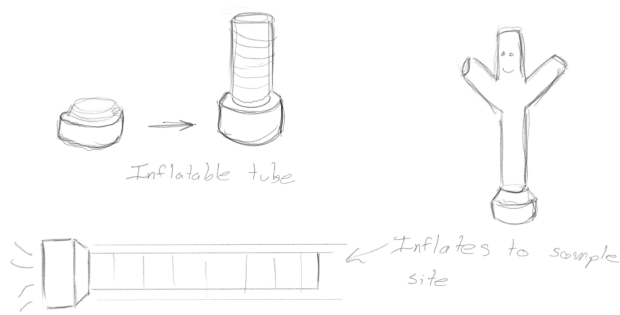

Figure 2.1: Ideation 1 (SS1) - Inflatable Pipe Navigator (Tiam Morrow-Rogers)

The Extendable Rod is a compact design that can be extended to meet the required sampling location inside the tube. It may be done manually by gradually extending a portion of the device, which may be done manually (by hand) or automatically or electronically controlled.

Figure 2.2: Ideation 2 (SS1) - Extendable Rod (Eden Irwin)

Wheels allow a tool to reach the exact location on the desired sample site in the pressure tube by rolling along the length of the pressure tube.

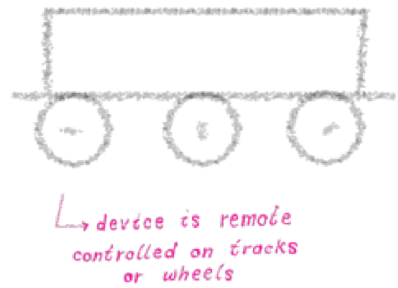

Figure 2.3: Ideation 3 (SS1)- Wheels (Eden Irwin, Tiam Morrow-Rogers, Isaac Pinarski, Radhe Pandey)

The Flywheel Launcher allows a tool to reach the desired sample location in the pressure tube by launching the tool payload down the tube's length.

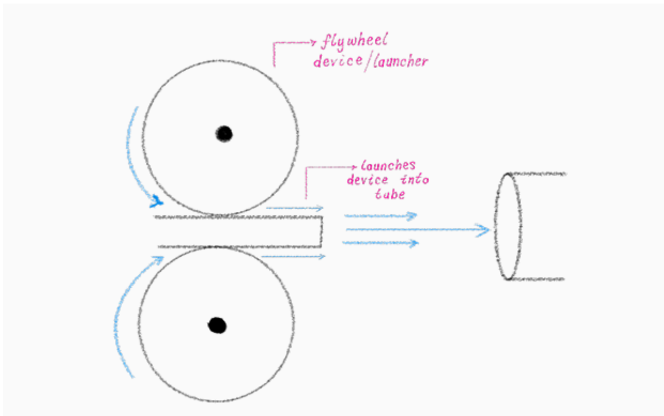

Figure 2.4: Ideation 4 (SS1) - Flywheel Launcher (Tiam Morrow-Rogers)

The inchworm transverse allows the tool to reach the desired sample site in the pressure tube by inflating one portion to become stuck in the tube and then using a linear actuator to push the rest of the body. Then, the other section can deflate, allowing the previous side to inflate and pull in.

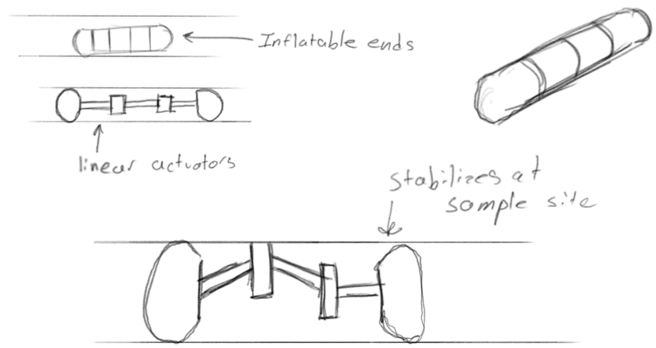

Figure 2.5: Ideation 5 (SS1)- Inchworm Transit (Tiam Morrow-Rogers)

Articulating links of a set length allows a device to roll out in a directionally controlled manner. The links can roll into a compact form, and the tool head can be attached to the end of each link for sampling. This device is also reasonable to move if the links don't articulate in the expected way.

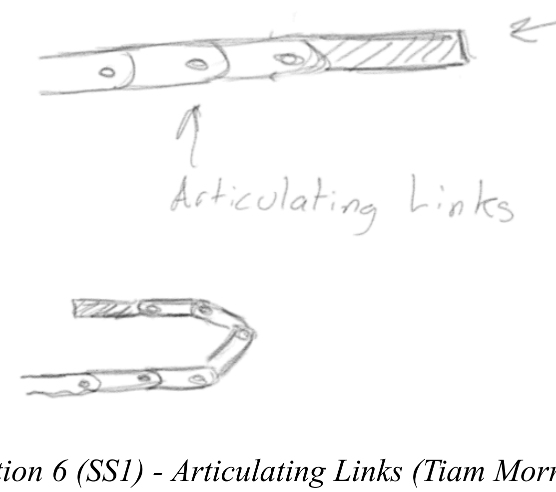

Figure 2.6: Ideation 6 (SS1) - Articulating Links (Tiam Morrow-Rogers)

A spool with a spring-action spike allows the device to be dropped to the necessary location, and the spikes are deployed, locking in the correct position. Pulling on the device with the spool retracts the spike and allows retrieval of the device. This method only works for vertically oriented pressure tubes.

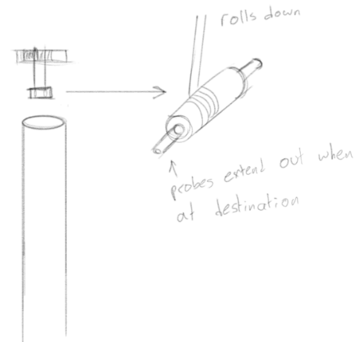

Figure 2.7: Ideation 7 (SS1) - Rolling Rope (Tiam Morrow-Rogers, Eden Irwin)

Table 2.1: Benefits and Drawbacks of Subsystem 1 Ideations

Ideation	Benefit	Drawback
Inflatable Pipe Navigator	It can reach the exact location of the sample site.	The attached tooling might be challenging to control, and the pressure needed for the inflatable to function might be challenging to achieve
Extendable Rod (Manual)	It can be easily controlled to meet the location of the sample site.	The performance of this method may take more time. The tool may also weigh a lot, and the device attached to the end must be light.
Wheels/Tracks	A “quick” method to reach the sample site. Easy to implement.	The device may lose power during travel.
Flywheel Launcher	A “quick” method to reach the sample site.	Errors could cause damage to the pressure tube or device. It may require consistent maintenance and tuning.
Inchworm Transverer	Meticulous movements that ensure stable travel along the inner tube walls. Can provide stability for tooling to perform sampling.	The performance of this method will be slow as the device's operational complexity is high. Many steps to move a distance.
Articulating Links	It has an accurate distance and is easy to retrieve from the pressure tube.	There are lots of points for mechanical failure. Difficult to clean and maintain.
Rolling Rope	It can be easily controlled to meet the location of the sample site.	The effectiveness of this method is for a vertical orientation of the tube.

Subsystem 2: Sampling

The One-Sided Scrape simply scrapes the side (or top of the inner tube wall). The scraping head moves horizontally (back and forth) against the tube wall to obtain the sample as the device is pulled back (towards the operator).

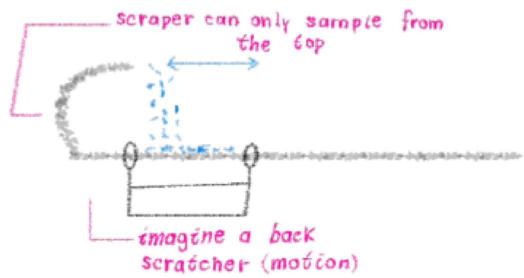

Figure 2.8: Ideation 1 (SS2) -One-Sided Scrape (Eden Irwin)

The Roof Scraper collects a sample by traversing the pressure tube while scraping the length of the sample site.

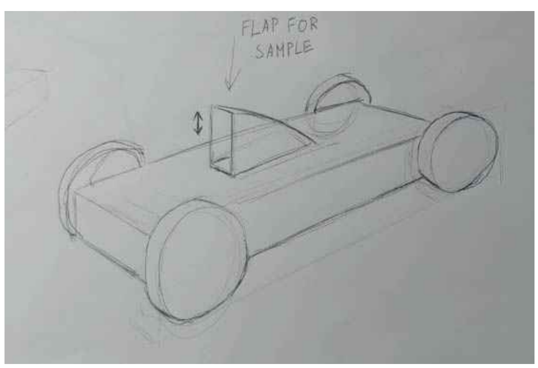

Figure 2.9: Ideation 2 (SS2) - Roof Scraper (Tiam Morrow-Rogers)

The Multi-Drill collects a sample by drilling the sample site while flushing the site with water. The vacuum tube is then used to suck in the sample solution.

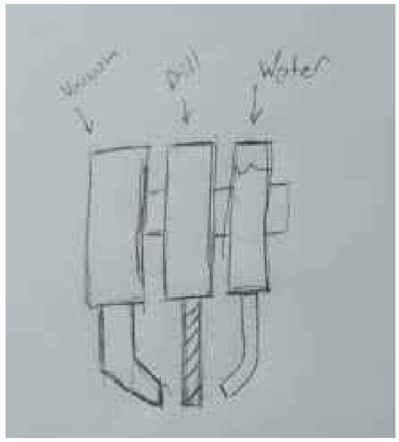

Figure 2.10: Ideation 3 (SS2) -Multi-Drill (Tiam Morrow-Rogers, Radhe Pandey, Ang Li)

The Electrolyte Swab would obtain the sample by coating the swab with an electrolyte solution (i.e. acid) strong enough to extract the necessary sample size from the inner tube wall.

Figure 2.11: Ideation 4 (SS2) - Electrolyte Swab (Eden Irwin)

The blade scraper collects a sample by repeatedly scraping the sample site. Teeth and a serrated edge are used to make the scraping more effective.

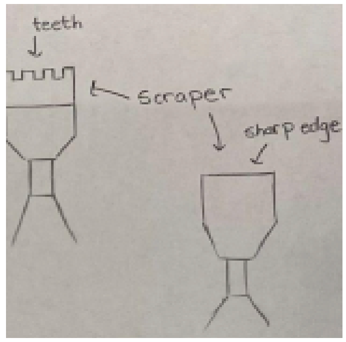

Figure 2.12: Ideation 5 (SS2) - Blade Scraper (Isaac Pinarski, Radhe Pandey)

The Sample Sander collects a sample by rotating a cylinder with a rough surface within the sample site.

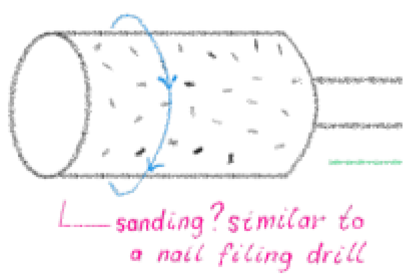

Figure 2.13:Ideation 6 (SS2) - Sample Sander (Eden Irwin)

Table 2.2: Benefits and Drawbacks of Subsystem 2 Ideations

Ideation	Benefit	Drawback
One-Sided Scraper	Low complexity and easy to maintain.	A longer process is needed as the sample is scraped upon second motion (i.e., pulled towards, not away).
Roof Scraper	Utilizes the component of another subsystem to achieve results.	Undetermined allowable extraction site distance. (Can it start before 15ft? Can it end after 15ft?)
Multi-Drill	The sample is more straightforward to collect and gives more precise control.	Requires a lot of components.
Electrolyte Swab	A simple extraction process. (i.e., not a lot of components)	Caution must be considered when dealing with chemicals, which may affect the sample quality.
Blade Scraper	A practical and straightforward method of retrieving the sample.	The blade can break, affecting sample quality and the material collected.
Sample Sander	A sample can be collected from a larger surface area.	This method would require consistent replacement of the sanding surface to maintain effectiveness. Difficult to clean.

Subsystem 3: Containment

The Sampling Box contains a sample using a flap. Once an adequate sample has been placed inside the box using a sampling tool, the flap closes, sealing and containing the sample.

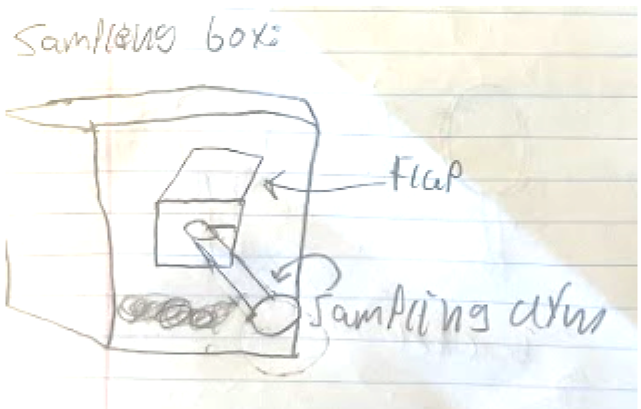

Figure 2.14: Ideation 1 (SS3) - Sampling Box (Eden Irwin, Tiam Morrow-Rogers, Isaac Pinarski, Radhe Pandey)

The Lens Aperture's fins are oriented, so when the top(lid) twists, it closes the container. Twisting the container again will open the lens, allowing testers to remove the material controllably.

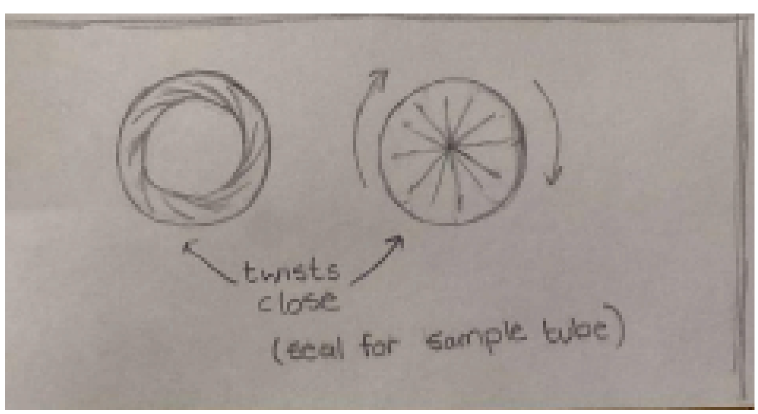

Figure 2.15: Ideation 2 (SS3) - Lens Aperture (Radhe Pandey)

The Vial is directly implemented in the design. Once the appropriate sample is achieved, the sealed vial can be extracted from the device (via a flap) and transported for testing. The vial becomes sealed upon a rapid, forceful motion to allow a cap to attach to the vial.

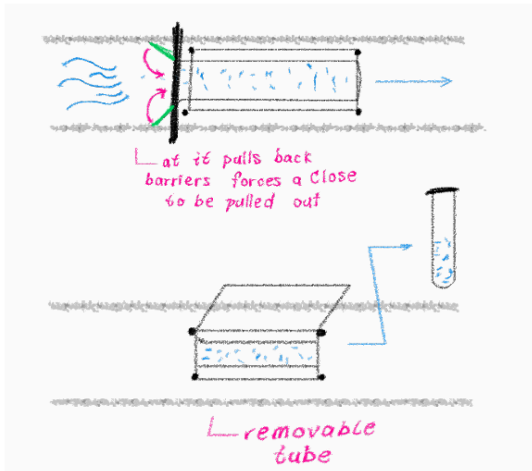

Figure 2.16: Ideation 4 (SS3) - Vial (Eden Irwin)

This idea allows the sample to be sealed in a clam-like enclosure and then dropped into a box. This box would enable the clamshell to remain closed and let the user remove the material without excessive direct contact.

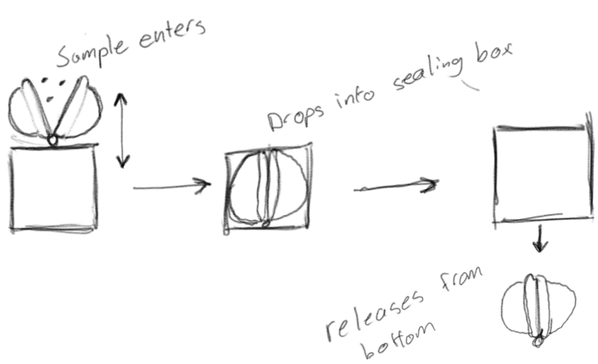

Figure 2.17: Ideation 5 (SS3) - Clamshell (Tiam Morrow-Rogers)

Table 2.3: Benefits and Drawbacks of Subsystem 3 Ideations

Ideation	Benefit	Drawback
Sampling Box	A simple configuration to contain the sample without contamination. Low complexity.	Sealing the flap of the sample box has not been entirely considered.
Lens Aperture	The retracted fins can hide inside the body, reducing exposure to contaminated material.	In high complexity, a failing fin can cause exposure to the sample.
Vial	A simple configuration to contain the sample without contamination.	The considered method of achieving a sealed vial has not been entirely considered.
Clamshell	Low complexity, easy-to-replace solution. After the container is sealed, the device can process the exterior to reduce contamination.	Opening the clamshell is potentially a contamination risk.

Subsystem 4: Failsafe

The reel is a retrieval device that extracts the entire device from the tube. The line of the reel is directly attached to the device. Its function is similar to a fishing rod or a winch.

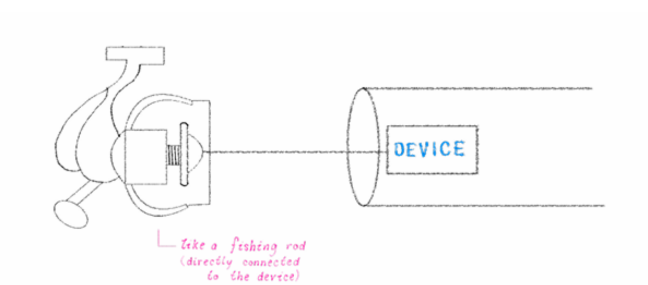

Figure 2.18: Ideation 1 (SS4) - Reel (Tiam Morrow-Rogers)

The Suction tool (i.e., vacuum) can ideally suction the device out of the tube. The suction would allow the device to reach the inlet of the tube to be further extracted.

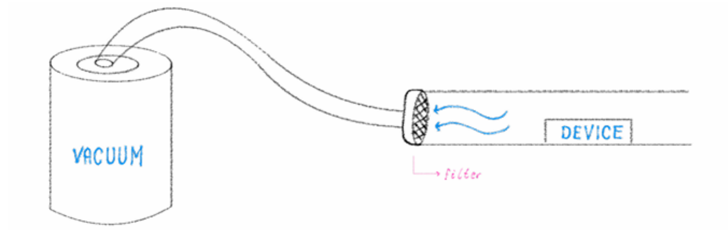

Figure 2.19: Ideation 2 (SS4) - Suction (Tiam Morrow-Rogers)

The device can be manually extracted from the tube. This method would only apply to a design that elongates from the tube's inlet to the sample's extraction site.

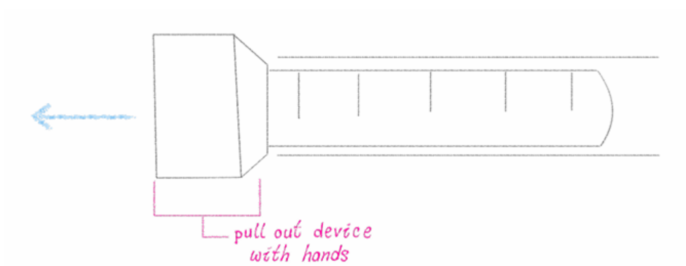

Figure 2.20: Ideation 3 (SS4) - Manual Extraction (Tiam Morrow-Rogers)

The device can be connected to a rope/or string to be pulled outside the tube after the sample extraction or in the event of device failure.

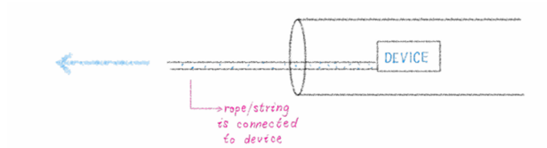

Figure 2.21: Ideation 4 (SS4) -Rope/String (Tiam Morrow-Rogers)

Table 2.4: Benefits and Drawbacks of Subsystem 4 Ideations

Ideation	Benefit	Drawback
Winch	This method can be manually or automatically controlled. It is very easy to maintain. It is an improved version of Rope/Chain.	The rope or chain of the winch could be a contamination risk. Difficult to clean.
Suction	Ideal for modular and lightweight designs with consistent force.	This method may only be effective for light devices where the vacuum pressure exceeds the restricting forces on the device.
Manual Extraction	Direct contact with the device to extract from the pressure tube.	This method is only applicable for devices elongating to the tube's inlet.
Rope/Chain	A simplistic method for removal of the device from the tube.	The inquiry of manual operation may lead to a slower extraction and inconsistent control.

3.0 Concept Refinement & Selection

3.1 Team Discussion & Refinement

The team conducted a detailed discussion to categorize, define, and refine the developed ideations. Each member presented their ideations for each subsystem: movement, sampling, containment, and failsafe.

The sketches presented a beneficial visual aspect to grasp the contents of the ideation and a description to support the vision. We noticed similar features on several as we presented and compared these ideations. The team organized all of the ideations for each subsystem to determine fully functional solutions by combining ideations from each respective subsystem.

Further analysis was required to evaluate the functionality of the concepts as the subsystem ideations were combined. The ideations have set a basis for implementing the design, but have been refined to suit the fully developed concept.

As a result, the team has developed four fully functional solutions by mixing and matching the subsystem ideas. Considering the client's desire to see novel solutions, some concepts were made without feasibility, group member ability, and cost.

Table 3.1: Concept Development from Subsystem Ideations

Concept Name	Movement	Sampling	Containment	Failsafe
Sample Subway	Wheels	Scraping	Lens Aperture	Reel
Pipe Worm	Inch Worm	Sander	Suction	Vacuum
Pneumatically Deployable Sample Tool	Inflatable Tube	Mechanical Arm	Clamshell	Deflate and Pull
Flappy Flywheel Flinger	Flywheel Launcher	Flap Scraper	Flap Closure	Reel

3.2 Concept Development

Concept 1: Sample Subway

The Sample Subway uses wheels to guide itself down the 15ft tube. A rope is attached to one end of the device, and with the help of a crank, the device can be led down the tube and retracted back out. A scraper is attached to the top of the device to shed material from the inside of the tube. An opening in the device collects the sampled material. The sample is secured within a container that twists closed, preventing contamination.

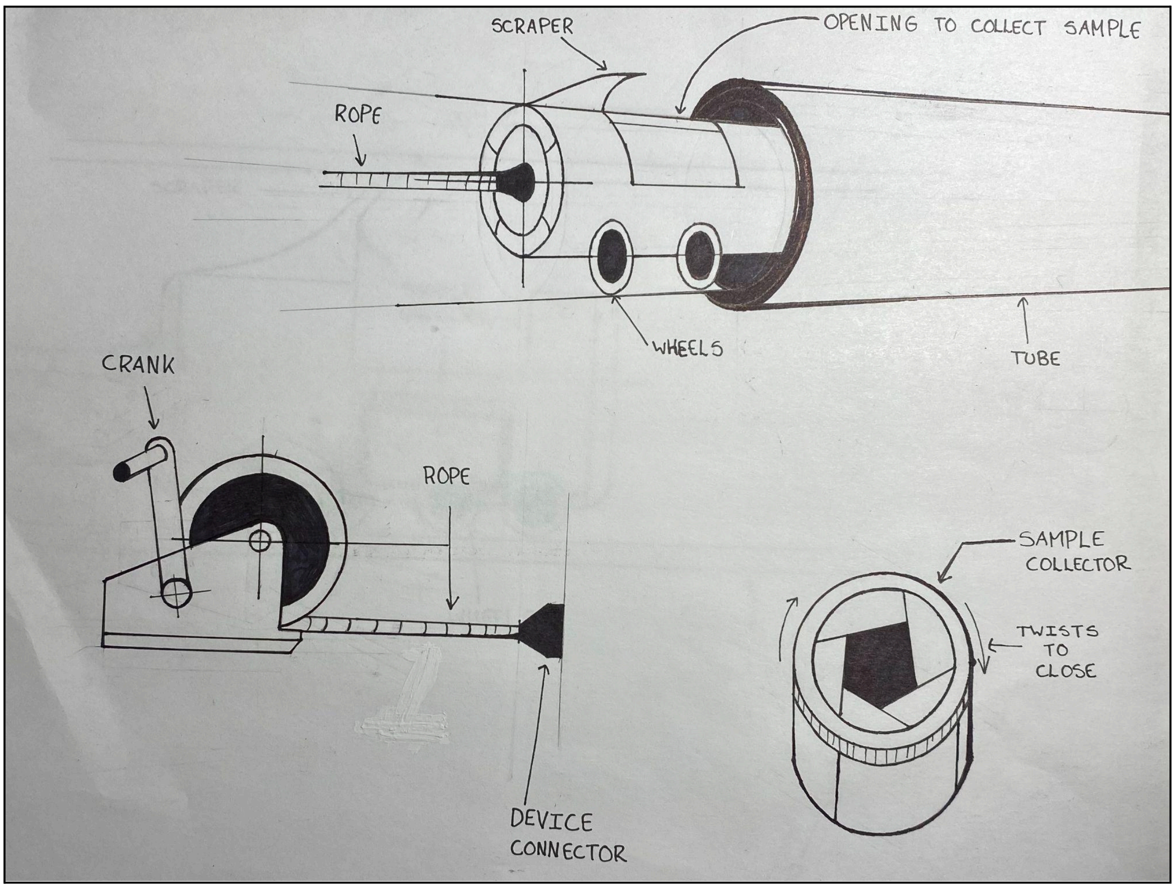

Figure 3.2-1: Sample Subway (1) (Radhe Pandey)

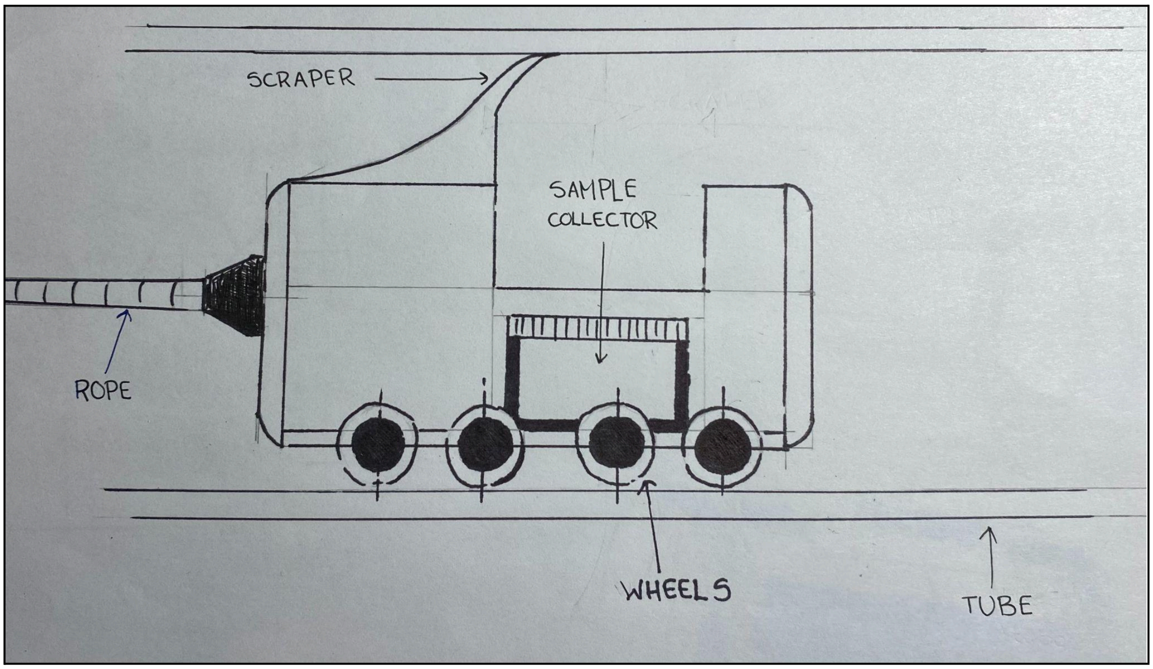

Figure 3.2-2: Sample Subway (2) (Radhe Pandey)

Concept 2: The Pipe Worm

The Pipe Worm uses inflatable end caps and a linear actuator to push and pull itself to the sampling site. It can then position itself to sand the surface of the pipe and suck in the material it removes. The device can then navigate back out of the tube, or it can be sucked out by a larger vacuum.

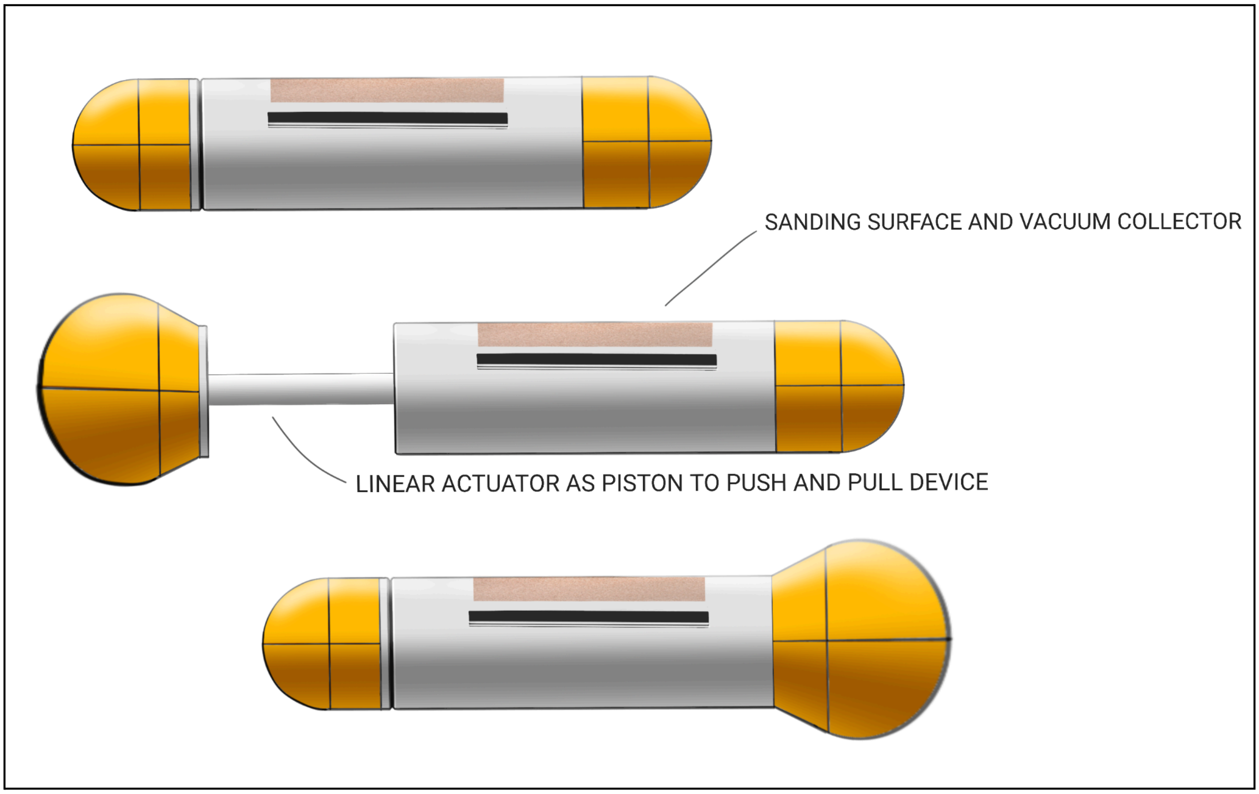

Figure 3.2-3: Pneumatically Deployable Sample Tool (Tiam Morrow-Rogers)

Concept 3: Pneumatically Deployable Sample Tool

The pneumatically deployable sample tool combines the Inflatable Tube from the movement subsystem, the mechanical arm from the sampling subsystem, and the clamshell containment method. It has an integrated failsafe. The gray cylindrical component inflates the orange tube by expanding the segments and pushing the tool head, depicted in blue below, to the sample site. The tool head can then rotate using the mechanical arm to deploy drilling or scraping tools and collect with the containment solution on the other side.

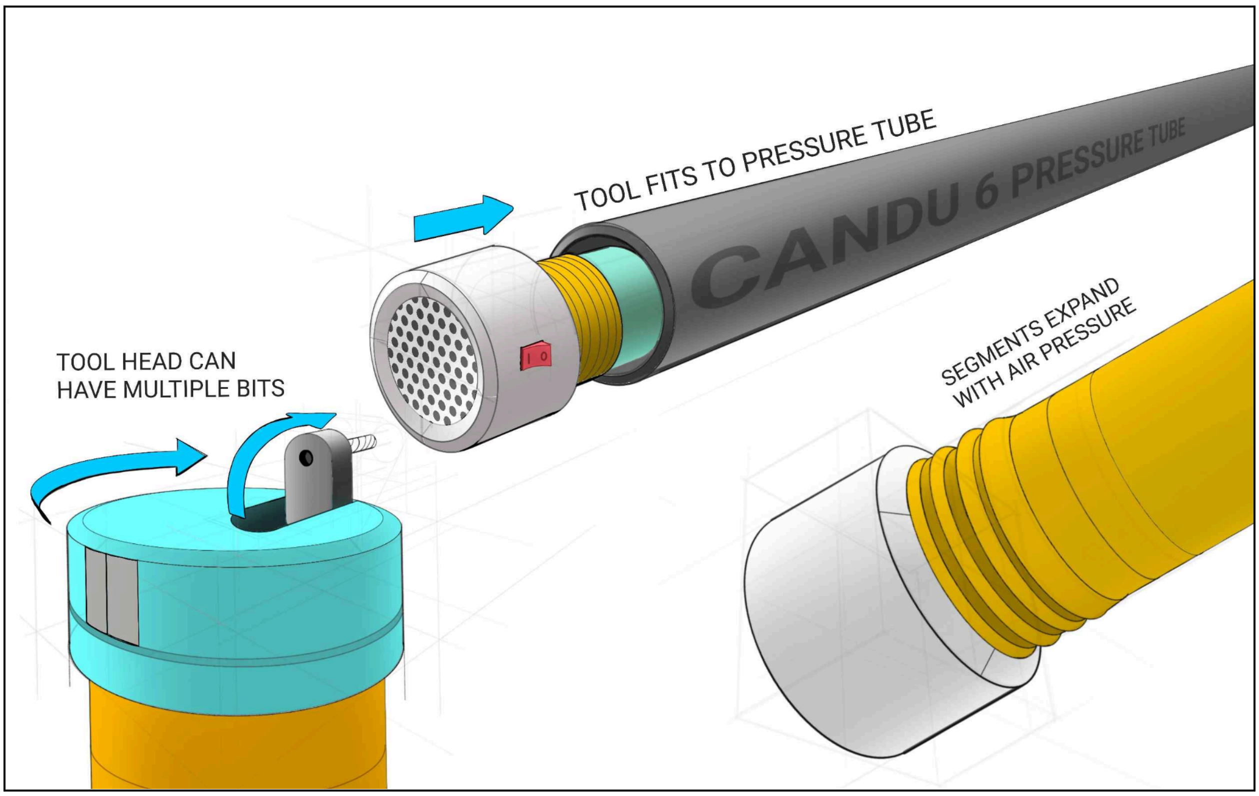

Figure 3.2-4: Pneumatically Deployable Sample Tool (Tiam Morrow-Rogers)

Concept 4: Flywheel Propelled Scrape Tool

The flywheel catches the scraping tool, launching it down the pipe. As the tool scrapes against the pipe wall, it collects the sample. The device can then be reeled back in. The friction of the scraper is multi-purpose in this scenario.

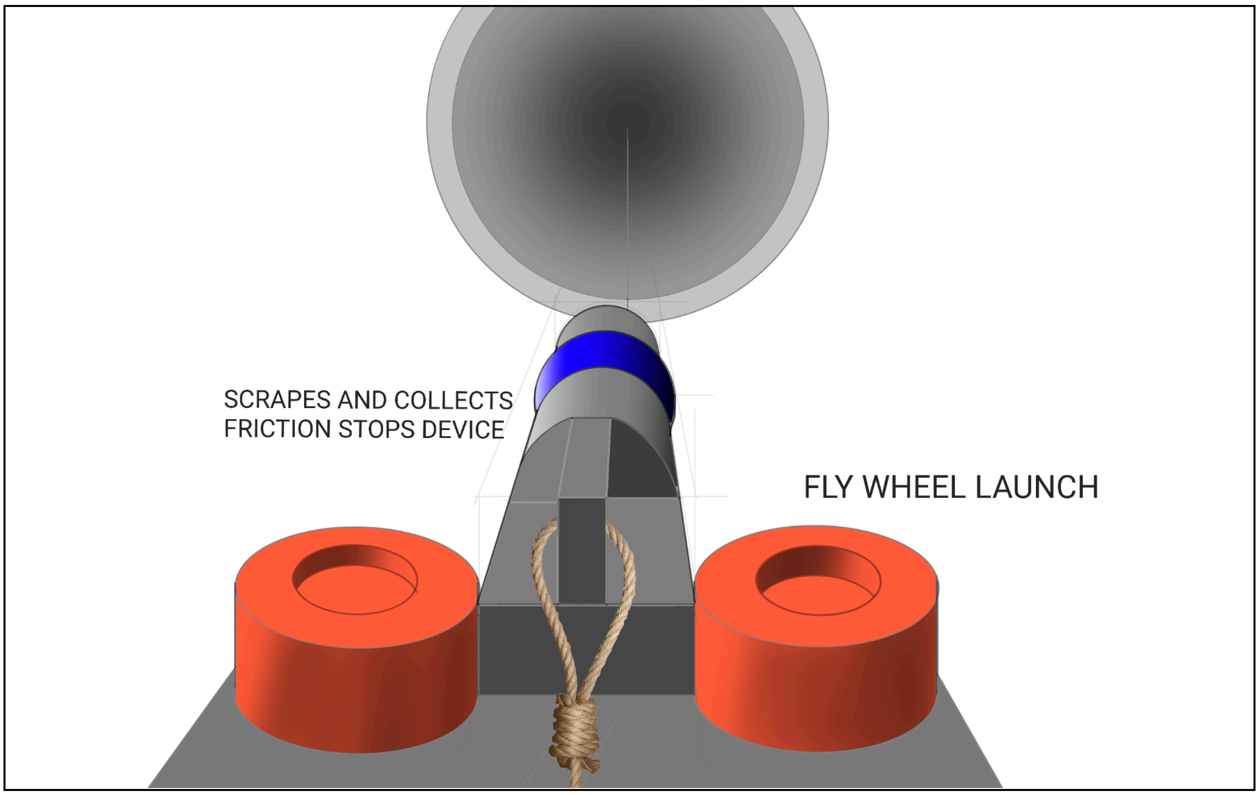

Figure 3.2-5: Flywheel Propelled Scrape Tool (Tiam Morrow-Rogers)

3.3 Concept Evaluation

Using the following design criteria, a selection matrix will be made to consider the benefits and drawbacks of each solution through a focused lens.

Table 3.2-1: Design Criteria

Need	High Priority	Design Criteria
The sample is contained to limit exposure	🛟SAFETY🛟	Thickness (mm) Material (Density)
Design can be completely removed	🛟SAFETY🛟	Maneuverable
The sample collected can be varied		30mg - 80mg sample collection
The design can report the current stage in its process		Display Feedback
Should be modular/lightweight		Weight (lbs) size (cm^3)
Necessary power sources should be included		Batteries/Compressed Air/Rat
Is a novel creative solution		Unique (by comparison)
The device is reasonably fast		Time (s)
The device is able to reach the sample site in the tube		Travel Distance (ft)
The device can fit inside the tube		Diameter (in.)

Presented in the depicted table, the team has evaluated the capabilities of each concept based on benefits, drawbacks, and possibilities of difficulty. This has been developed into an evaluation scale from 1-3, as shown below.

Table 3.3-2: Evaluation of Concepts Based on Design Criteria

Evaluation Scale	1	2	3

Concept Number → Design Criteria ↓	Concept 1	Concept 2	Concept 3	Concept 4
Containment	Easy to implement a controllable solution	The sanding pad would be highly contaminated	The multi-tool head could implement a better subsystem	The flap could break off
Removable	Dependent on reel in the event of motor failure or power loss	Could get stuck	The air pump can be run in reverse to pull the inflatable back in	High risk of damage or loose parts
Modular/Lightweight	Depends on materials.	Depends on materials.	The device packs down small	The launcher may be heavy
Creative Solution	It's a drone.	Is a creative solution (via inchworm)	Is a creative solution (via an inflatable tube)	Is a creative solution (via flywheel launcher)
Reasonably Fast	Motors provide good speed and torque	Many steps for movement slow down speed	Inflates faster than a twin-size air mattress	Very high-speed launch
Fits Inside Tube	Need to consider motor size	It might not be able to be sized appropriately	Powered components can be outside tube	Can launch a smaller device
Complexity	Many resources for similar designs	It has a lot of mechanical operations for a single action	Combining the functions of all components.	Combining the functions of all components.
TOTAL	**16**	**11**	**19**	**15**

3.4 Final Concept Selection

Based on the rating scheme depicted in the evaluation of concepts, each design solution has been assessed against the prioritized design criteria, which have been scored accordingly to determine the "best" solution based on meeting design criteria.

Through this evaluation process, the best global solution is the *Pneumatically Deployable Sample Tool*, Concept 3. After consideration, the air pressure required for the device is not significant. Standard air mattress pumps are more than capable of providing the necessary air pressure. It is also assumed that very few have considered this approach. The team believes that the tool head could utilize many different solutions for sampling. There is an opportunity to test different sampling subsystems with this tooling delivery system.

4.0 Client Meeting 2 Preparation

4.1 Presentation Plan

The team must prepare a presentation for the second client meeting. During this time, we will present the steps taken within each stage of the Design Thinking process to validate and receive feedback in any aspect possible, as outlined in Table 4.1.

Table 4.1: Overview of Slide Content

Title Page	Group 16: Sample Snatchers - Client Presentation	1 slide
Introduction	Introducing the team members	1 slide
Empathize	Interpreted Need Statements	1 slide
	Hierarchy of Needs	1 slide
Problem Statement	Problem Statement	1 slide
Define	Design Criteria	1 slide
	Technical & User Benchmarking	2 slides
	Engineering Design Specifications (Target Specifications)	3 slides
Ideate	Subsystems (4)	1 slide
	Ideations of Subsystems	4 slides
	Development of Concepts (and Explanation)	4 slides
Questions	List of Questions	1 slide

This client presentation is essential for the project's progress. The team must meticulously organize the content to ensure a structured delivery that addresses all key aspects to facilitate feedback. We will interact with the client throughout the presentation to receive confirmation or comments during that phase. As the generated concepts are driven by prior work, validating these findings (i.e., the problem statement) is crucial. During this session, we will consider any suggestions the client may have. We can explicitly request feedback or interpret their language during the presentation.

4.2 Client Questions

Furthermore, the team has generated questions for the client regarding target specifications and any specific design clarifications regarding the project scope. This list will be further developed for the completed presentation.

- Can sampling start (or end) before (or after) 15ft? (Explain corresponding ideation)
- How long should the whole extraction process take (i.e. operation time)?
- Do we need to consider and design the external system (i.e. BRIMS)? Can we have direct access to the pressure tube?
- Is the pipe schedule known? (We might need to evaluate alternative methods for scraping on a larger surface area)
- Request clarification regarding the stage reporting of the process (i.e. digital screen). What feedback should the device display (i.e. weight, distance in the tube)?
- Do any of our ideations catch your eye more than our concepts?

5.0 Conclusion

5.1 Concept Selection for Client Meeting

The ideation stage provided the Sample Snatchers with a variety of ideations. The ideations were placed into four main subsystems: movement, containment, sampling, and failsafe. The ideations were then reviewed, and four concepts were formed by taking ideation from each subsystem and assimilating them into one concept. After creating Table 3.2-2 and grading each concept against relevant design criteria, the ***Pneumatically Deployable Sample Tool*** is chosen for the Sample Snatchers prototype. As client feedback is received and reviewed, these concepts will be modified to fit the criticism.

The group plans to reconvene on Monday afternoon to complete the slides for the client presentation. More questions will likely be developed in this timeframe as well. Further detailing of slide content will also be inspected.

5.2 Trello: Task Plan Update

In Trello, the team's task board has been updated to include changes in task duration, missing tasks, task responsibilities, and further details to maintain effective project management.

Please refer to the following link to review updates:
https://trello.com/invite/b/67846e4e793783740d87aed9/ATTIbb8224a0c532547c0a69acd41093c7c7A4D60EC7/gng1103-project-sample-snatchers

Additional table on the following page.